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Summary

Medical instruments help in correct diagnosis and treatment to observing patients, examining organs, making measurements and administering medication. Today's medical instruments are considerably more complicated and diverse, primarily because they incorporate electronic systems for sensing, transducing, manipulating, storing, and displaying data or information. The modern electronic stethoscope is a high precision instrument used to hear with clarity the internal sounds of the human body even in a noisy environment or through the heavy clothing of the patient. The sphygmomanometer is a device to diagnose pathological conditions such as hypertension and hypotension. A glucometer is a simple and portable medical device used to find out the approximate levels of blood glucose. Autoanalyser is a computer controlled device and used to quick estimation of various bio-chemical substances like glucose, urea, cholesterol, enzymes and other proteins present in body fluids.

Imaging instruments produce diagnostic images and help in the detection of the diseases or defects in an organ at a very early stage, when there are no clinical manifestations or the disease is undetected by the conventional methods of investigations. Then proper therapeutic or surgical measures can be adopted. An EEG is a test to evaluate the electrical activities of the brain. X-Rays are a form of electromagnetic radiation to visualize the internal structures of a patient. Ultrasound waves are used to image the foetus at different stages of pregnancy used to hear foetal heart sound, blood flow, diagnose the damages in heart and diagnosis of tumours, gall stones, kidney stones, and obstructions in the genital tracts.

Glossary

Berger's waves – are neural oscillations in the frequency range of 7.5–12.5 Hz arising from synchronous and coherent (in phase or constructive) electrical activity of thalamic pacemaker cells in humans.

Blood pressure – (BP) is the pressure exerted by the circulating blood against the walls of blood vessels.

Electromagnetic (EM) radiation – Electromagnetic radiation is a form of energy that is all around us and takes many forms, such as radio waves, microwaves, X-rays and gamma rays.

Hypotension – Hypotension is abnormally low blood pressure.

Incubator – is a device used to grow and maintain microbiological cultures or cell cultures. and maintains optimal temperature, humidity and other conditions such as the carbon dioxide (CO2) and oxygen content of the atmosphere inside.

Manometer – is an instrument for measuring the pressure acting on a column of fluid, consisting of a U-shaped tube of liquid in which a difference in the pressures acting in the two arms of the tube causes the liquid to reach different heights in the two arms.

Oscilloscope – It is a laboratory instrument commonly used to display and analyze the waveform of electronic signals. In effect, the device draws a graph of the instantaneous signal voltage as a function of time.

Piezoelectric Effect – It is the ability of certain materials to generate an electric charge in response to applied mechanical stress.